Noah Wolfe

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Education

2022– **Ph.D.**, Physics, Massachusetts Institute of Technology
2018–22 **B.S.**, Physics & **B.S.**, Mathematics, North Carolina State University

Research Experience

2024– **Junior Investigator**, NSF Institute for Artificial Intelligence and Fundamental Interactions (IAIFI)

2022– **Graduate Research Fellow**, LIGO Laboratory, MIT
Advisor: Salvatore Vitale

2021 **Summer Undergraduate Research Fellow**, LIGO Laboratory, Caltech
Advisors: Colm Talbot, Alan Weinstein

2018–22 **Undergraduate Research Assistant**, Dept. of Physics, NC State
Advisor: Carla Fröhlich

Awards, Honors, & Funding

2025 MIT Dept. of Physics & Plasma Science and Fusion Center travel and research funding ($10,000) for the 2026 Geophysical Plasma Observation Expedition
2022 Henry W. Kendall (1955) & La Gattuta Fellowships, Dept. of Physics, MIT
2022 National Science Foundation Graduate Research Fellowship
2022 Senior Award for Outstanding Research, College of Sciences, NC State
2022 McCormick Award for Undergraduate Research, Dept. of Physics, NC State (co-recipient)
2022 Yusor Abu-Salha Award for Outstanding Community Engagement, Dept. of Mathematics, NC State
2021 Astronaut Scholarship
2021 1st Place, McCormick Undergraduate Physics Research Symposium, Dept. of Physics, NC State
2020 Goldwater Scholarship
2020 NC State Sustainability Fund Grant ($5,820) for a campus-wide particulate matter sensor network
2020 NC State Office of Undergraduate Research Award ($2,000) for research at the Institute for Advanced Study (canceled due to COVID-19)
2020 Phi Beta Kappa
2020 Math Honors Program, Dept. of Mathematics, NC State
2018–22 Park Scholarship, NC State
Highly selective, full merit scholarship recognizing outstanding scholarship, leadership, service, and character
2018–22 University Scholars Program, NC State

Publications

ORCID

J1. Plunkett, C., Heinzel, J., **Wolfe**, N. E., Gerosa, D. & Vitale, S. Spinformation: do gravitational-wave events carry information beyond effective spins? *preprint*. arXiv: 2610.18973 [gr-qc] (Oct. 2026).

J2. **Wolfe**, N. E., Hussain, A., Heinzel, J. & Vitale, S. Binary-black hole spin population results may be driven by prior degeneracies. *preprint*. arXiv: 2609.10753 [gr-qc] (Sept. 2026).

J3. **Wolfe**, N. E., Vitale, S. & Zevin, M. No Model-independent Evidence for a Peak in Binary Black Hole Spin (Mis)alignments. *Astrophys. J. Lett.* **1007,** L45. arXiv: 2605.05300 [astro-ph.HE] (Aug. 2026).

J4. Mould, M., Heinzel, J., Álvarez-López, S., Plunkett, C., **Wolfe**, N. E. & Vitale, S. Measurement prospects for the pair-instability mass cutoff with gravitational waves. *Phys. Rev. D* **113,** 103021. arXiv: 2602.11282 [astro-ph.HE] (May 2026).

J5. **Wolfe**, N. E., Mould, M., Heinzel, J. & Vitale, S. Reducing systematic uncertainties in gravitational-wave population analyses by increasing the detection threshold. *Phys. Rev. D* **113,** 103025. arXiv: 2510.06220 [gr-qc] (May 2026).

J6. **Wolfe**, N. E., Mould, M., Veitch, J. & Vitale, S. Neural Bayesian updates to populations with growing gravitational-wave catalogs. *Phys. Rev. D*. arXiv: 2602.20277 [astro-ph.IM] (Feb. 2026). In press.

J7. De Lorenci, V. A., Kaiser, D. I., Peter, P., Ruiz, L. S. & **Wolfe**, N. E. Gravitational wave signals from primordial black holes orbiting solar-type stars. *Phys. Rev. D* **112,** 063063. arXiv: 2504.07517 [gr-qc] (Sept. 2025).

J8. Mould, M., **Wolfe**, N. E. & Vitale, S. Rapid inference and comparison of gravitational-wave population models with neural variational posteriors. *Phys. Rev. D* **111,** 123049. arXiv: 2504.07197 [astro-ph.IM] (June 2025).

J9. **Wolfe**, N. E., Vitale, S. & Talbot, C. Too small to fail: characterizing sub-solar mass black hole mergers with gravitational waves. *JCAP* **2023,** 039. arXiv: 2305.19907 [astro-ph.HE] (Nov. 2023).

J10. **Wolfe**, N. E., Fröhlich, C., Miller, J. M., Torres-Forné, A. & Cerdá-Durán, P. Gravitational Wave Eigenfrequencies from Neutrino-driven Core-collapse Supernovae. *Astrophys. J.* **954,** 161. arXiv: 2303.16962 [astro-ph.HE] (Sept. 2023).

J11. **Wolfe**, N. E., Talbot, C. & Golomb, J. Accelerating tests of general relativity with gravitational-wave signals using hybrid sampling. *Phys. Rev. D* **107,** 104056. arXiv: 2208.12872 [gr-qc] (May 2023).

J12. Meskhi, M. M., **Wolfe**, N. E., Dai, Z., Fröhlich, C., Miller, J. M., Wong, R. K. W. & Vilalta, R. A New Constraint on the Nuclear Equation of State from Statistical Distributions of Compact Remnants of Supernovae. *Astrophys. J. Lett.* **932,** L3. arXiv: 2111.01815 [astro-ph.HE] (June 2022).

J13. Ghosh, S., **Wolfe**, N. & Fröhlich, C. PUSHing Core-collapse Supernovae to Explosions in Spherical Symmetry. V. Equation of State Dependency of Explosion Properties, Nucleosynthesis Yields, and Compact Remnants. *Astrophys. J.* **929,** 43. arXiv: 2107.13016 [astro-ph.HE] (Apr. 2022).

J14. Curtis, S., **Wolfe**, N., Fröhlich, C., Miller, J. M., Wollaeger, R. & Ebinger, K. Core-collapse Supernovae: From Neutrino-driven 1D Explosions to Light Curves and Spectra. *Astrophys. J.* **921,** 143. arXiv: 2008.05498 [astro-ph.SR] (Nov. 2021).

Talks

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T1. * Joint Princeton–IAS Gravitational Wave Group Meeting, Princeton University (Princeton, NJ, USA)
What we do and don't (yet) know about binary-black hole spins. Scheduled. Nov. 2026.

T2. * CIERA, Northwestern University (Evanston, IL, USA)
What we do and don't (yet) know about binary-black hole spins. Scheduled. Oct. 2026.

T3. * Adler Planetarium (Chicago, IL, USA)
Geophysical Plasma Observation Expeditions. Scheduled. Oct. 2026.

T4. * Astronomy Chalk Talk, Dept. of Astronomy & Astrophysics, University of Chicago (Chicago, IL, USA)
What we do and don't (yet) know about binary-black hole spins. Scheduled. Oct. 2026.

T5. * STRAND Seminar, UC San Diego (San Diego, CA, USA)
Neural Bayesian updates to populations with growing gravitational-wave catalogs. May 2026.

T6. * AstroAI, Harvard & Smithsonian Center for Astrophysics (Cambridge, MA, USA)
Neural Bayesian updates to populations with growing gravitational-wave catalogs. Recording. Apr. 2026.

T7. Institute for Artificial Intelligence and Fundamental Interactions (IAIFI) (Cambridge, MA, USA)
Neural Bayesian updates to populations with growing gravitational-wave catalogs. Apr. 2026.

T8. American Physical Society April Meeting (Denver, CO, USA)
Neural Bayesian updates to populations with growing gravitational-wave catalogs. Mar. 2026.

T9. American Physical Society April Meeting (Minneapolis, MN, USA)
Too small to fail: Measurability of sub-solar mass compact objects with gravitational waves. ADS. Apr. 2023.

T10. OzGrav Collaboration-wide Meeting (Melbourne, VIC, Australia)
Relations for Neutron Star Structure with Gravitational-Wave Signals from Core-Collapse Supernovae. Slides. July 2022.

T11. McCormick Research Award Colloquium (Raleigh, NC, USA)
Physics in the Stellar Graveyard. Slides. May 2022.

T12. LIGO SURF 2021 Final Presentations (Pasadena, CA, USA)
Testing General Relativity with Gravitational Wave Signals. Slides. Aug. 2021.

T13. McCormick Undergraduate Research Symposium (Raleigh, NC, USA)
Characterizing Gravitational Wave Signals from Core-Collapse Supernovae. May 2021.

T14. UMass Dartmouth Physics Colloquium (Dartmouth, MA, USA)
Characterizing Gravitational Wave Signals from Core-Collapse Supernovae. Apr. 2021.

T15. McCormick Undergraduate Research Symposium (Raleigh, NC, USA)
Characterizing Gravitational Wave Signals from Core-Collapse Supernovae. Slides. May 2020.

Academic Advising

2024 Zeineb Mezghanni, MIT Summer Research Program (MSRP)
Now a Physics Ph.D. student at MIT!

Academic Service

Collaboration Service

2026– Analyst & reviewer, LIGO–Virgo–KAGRA (LVK) astrophysical populations papers
Contributing an analysis and write-up to the forthcoming LVK astrophysical populations paper, based on the approach I presented in arXiv:2510.06220

2023–25 Parameter estimation rota, LVK O4 observing run
Volunteered for several week-long on-call shifts monitoring parameter estimation during the observing run, ensuring analyses ran as expected

2023 Parameter estimation liaison, LVK search for sub-solar mass black holes (arXiv:2605.05444)
Expert on the paper writing team responsible for source parameter estimation of any significant sub-solar mass black hole candidates

Journal Reviewer

2026– Physical Review D

2026– The Astrophysical Journal

2026– Monthly Notices of the Royal Astronomical Society

Leadership & Community Service

2024–

Lead Organizer, MIT Geophysical Plasma Observation Expedition (featured in MIT News)
GPOE is an entirely graduate-student organized and led trip to Fairbanks, AK to study the aurora.

- **2025 expedition** (organized from Fall 2024): selected to join; designed and implemented firmware to run low-cost, all-sky cameras and magnetometers for multi-day observations, enabling 3D (tomographic) reconstruction of aurorae.
  - Developed an intensive two-day curriculum with the MIT Museum, leading 65 high school students from around the world in a design-thinking exercise to prototype the all-sky cameras we deployed.
- **2026 expedition** (organized from Fall 2025): led the expedition, defining the scientific program in collaboration with expedition members and organizing trip logistics (funding, flights, and lodging).
- Led candidate selection for the expedition team.
- Secured $10,000 in funding from MIT despite Fall 2025 budget cuts, including $3,500 from a gift by Prof. Andrea Ghez (MIT 1987) in honor of MIT Prof. Hale Bradt.
- Designing a planetarium show from our all-sky footage, to be presented at Boston-area schools in Fall 2026.

2024 — Math & science tutor, West End House Boys & Girls Club (Boston, MA)
Weekly math and science tutoring for local high school students

2020–22 — **President & Co-Founder**, Scivir
501(c)(3) nonprofit building low-cost mesh networks of particulate matter sensors to close air quality data gaps in rural North Carolina; led its organizational and technical vision and completed multiple small grant applications

2020 — Alternative Service Break, Hoonah, AK
Volunteered at the local school and Boys & Girls Club in a Tlingit-majority community to learn about social justice and education

2020 — Small-group leader, Triangle Youth Leadership Conference
Guided high school students through a two-day conference on leadership and community problem solving

2019–21 — **President**, Astronomy Club at NC State
Led the club in science outreach, including events in Raleigh and remote events during 2020–21

2018–22 — **Co-Chair**, Park Scholars Class of 2022 Legacy Committee
Led the class legacy project and a partnership with the Interfaith Food Shuttle on food insecurity and sustainability

Outreach Talks

2023	"Science in the Stellar Graveyard," Malden High School student science club, Malden, MA
2022	"Science in the Stellar Graveyard," North Carolina Science & Engineering Fair (virtual) Recording
2022	"The Second Lives (and deaths) of Stars," Martin GT Magnet Middle School, Raleigh, NC (virtual) Slides
2021	"The Deaths of Stars," Martin GT Magnet Middle School, Raleigh, NC (virtual)
2020	"The Deaths of Stars," Martin GT Magnet Middle School, Raleigh, NC (virtual)
2020	Solar astronomy tutorial for special-needs students, Athens Drive Magnet High School, Raleigh, NC
2020	Lunar & planetary astronomy tutorial, Millbrook Magnet Elementary School, Raleigh, NC
2020	Astronomy Days outreach booth, North Carolina Museum of Natural Sciences, Raleigh, NC

Last updated: October 16, 2026